Question 1: Career in Software Industry

Answer:

I would tell my friend that today, the software industry is open to everyone — not just people from big cities or top colleges. What matters most are **skills, practice, and consistency**, not background.

There are many roles available in IT such as:

- Software Developer (builds applications)
- Web Developer (creates websites)
- Mobile App Developer
- UI/UX Designer (designs user interface)
- Tester/QA Engineer (checks for bugs)
- Data Analyst

To get started, important skills include:

- Basic programming (Python, Java, JavaScript)
- Problem-solving ability
- Logical thinking
- Communication skills
- Learning attitude

Many people learn online and get jobs through projects and skills, so anyone can succeed.

Question 2: Roles in Building a Mobile App

Answer:

To build a mobile app, a team of different people works together:

- **Product Manager** – decides what the app should do
- **UI/UX Designer** – designs how the app looks and feels
- **Frontend Developer** – builds what users see in the app
- **Backend Developer** – manages data, servers, and logic
- **Mobile Developer** – creates the app for Android/iOS
- **Tester (QA)** – checks for errors and bugs
- **DevOps Engineer** – deploys the app and manages servers

They work together step-by-step:

1. Plan the app
2. Design it

3. Develop it
4. Test it
5. Launch it

Question 3: Why SyntaxError Happens

Answer:

Computers do not understand human language — they only understand **machine language (0s and 1s)**.

When we write code in Python:

1. The code is converted into a form the computer understands.
2. If there is a mistake in writing (like missing brackets or wrong spelling), the computer cannot understand it.

A **SyntaxError** means:

- The code is not written in the correct format or grammar of the programming language.

It's like writing a sentence with wrong grammar — the meaning becomes unclear.

Question 4: Algorithm and Pseudocode

Answer:

Plain English Steps:

1. Start
2. Enter total bill amount
3. If bill > 500:
    - Calculate 10% discount
    - Subtract discount from total
4. Otherwise:
    - No discount
5. Show final bill
6. End

Pseudocode:

START
INPUT total_bill

```
IF total_bill > 500 THEN
   discount = total_bill * 0.10
   final_bill = total_bill - discount
ELSE
   final_bill = total_bill
END IF

PRINT final_bill
END
```

Question 5: How Website Loads

Answer:

When you type a website like www.flipkart.com:

1. **DNS (Domain Name System)**:
   - Converts the website name into an IP address.
2. **Request to Server**:
   - Your browser sends a request to the server using **HTTP/HTTPS**.
3. **Server Response**:
   - The server sends back website data (HTML, CSS, JavaScript).
4. **Browser Rendering**:
   - The browser reads the data and displays the webpage.

This entire process happens in seconds due to fast internet and optimized systems.

Question 6: Cloud Computing

Answer:

Cloud computing means using **internet-based servers instead of owning physical computers**.

Benefits for startups:

- No need to buy expensive hardware
- Pay only for what you use
- Easily scalable (can handle more users anytime)
- Fast and reliable

Example:
A food delivery app can store data on cloud platforms like AWS or Google Cloud instead of buying servers. Users access the app online, and all data is handled remotely.

Question 7: Compiler vs Interpreter

Answer:

- **Compiler**:
  - Converts entire code into machine language at once.
  - Faster execution after compilation.
- **Interpreter**:
  - Converts and runs code line-by-line.
  - Slower but easier to debug.

Real-life example:

- Compiler = Translating a whole book at once.
- Interpreter = Translating sentence-by-sentence while reading.

Question 8: SDLC Phases

Answer:

SDLC (Software Development Life Cycle) is the process of building software.

Phases:

1. **Planning** – Decide what to build
2. **Requirement Analysis** – Understand user needs
3. **Design** – Create system structure
4. **Development** – Write code
5. **Testing** – Check for errors
6. **Deployment** – Release to users
7. **Maintenance** – Fix bugs and update

Question 9: Frontend vs Backend

Answer:

- **Frontend**:
  - What users see (UI)

    - Example: buttons, images, layout
- **Backend**:
    - Behind-the-scenes logic
    - Manages database and server

Example:
In a shopping app:

- Frontend shows products
- Backend processes orders and payments

Both work together to make the app function.

Question 10: HTTP vs HTTPS

Answer:

- **HTTP**:
    - Not secure
    - Data is sent in plain text
- **HTTPS**:
    - Secure (encrypted)
    - Protects user data

Importance:
HTTPS is important for banking or payment websites because:

- Keeps passwords and card details safe
- Prevents hacking and data theft